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| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 06/19/2009 |
| **PI Name:** | Rood, Aaron M.S. | | |
| **Project Title:** | Lunar Health Monitor: A wearable system to monitor astronaut health status | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Operational and clinical research | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **ExMC**:Exploration Medical Capabilities | | |
| **Human Research Program Risks:** | (1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | rood@orbitalresearch.com | **Fax:** | FY |
| **PI Organization Type:** | INDUSTRY | **Phone:** | |
| **Organization Name:** | Orbital Research Inc. | | |
| **PI Address 1:** | 4415 Euclid Avenue, Suite 500 | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Cleveland | **State:** | OH |
| **Zip Code:** | 44103-3757 | **Congressional District:** | 11 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | SBIR Phase II |
| **Start Date:** | 05/30/2008 | **End Date:** | 05/29/2010 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Watkins, Sharmi1a | **Contact Phone:** | 281.483.0395 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Orbital Research has successfully demonstrated a dry electrode (no electrolyte or gel required) for heart rate and ECG monitoring. Preliminary data has indicated that the electrode can be incorporated into a garment that does not require an adhesive for successful data collection. Further preliminary data indicates that respiration rate can be deduced from the ECG waveform captured from the dry electrodes. The Orbital Lunar Health Monitor (LHM) will incorporate the patented dry electrode with other commercially available sensors in a garment resulting in fully wearable monitor of Astronaut Health. The data from each of the sensor is processed, yielding a comprehensive image of overall health. Environment sensors are also incorporated to detect longer term health threats. The LHM is comfortable and non-inhibiting. The Orbital Lunar Health Monitor is intended to enable continuous monitoring of health status of Astronauts both in intra- and extra-vehicular sorties.<br>POTENTIAL NASA COMMERCIAL APPLICATION(S): The Lunar Health Monitor will be a fully wearable health | | |

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| | monitor specifically proposed for continuous monitoring of Astronauts during intra- and extra-vehicular activity. The overall function of the system is not limited in any way to lunar missions. The wearable system could be used for physiologic monitoring during training exercises or continuous monitoring of day to day physiologic profile. The Lunar Health Monitor can easily be worn under flight gear or with civilian clothing, making the system completely versatile for any applications where continuous physiologic monitoring is needed. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Orbital Research believes the wearable Lunar Monitor System can be applied to health/lifestyle management, clinical assessments, drug trials and first responders. The continuous monitoring allows detailed understanding of health factors such as activity levels and duration and caloric demand. The health monitor allows verification of lifestyle management practices. Clinicians and health care practitioners can use the technology for out of hospital assessment, much like a Holter monitor is currently used. The Orbital Research technology will enable drug studies to monitor the chronic response and change over time. Finally, persons such as first responders or war fighters can use the monitor to assist in making life or death decisions. |
| **Task Progress:** | New project for FY2008. Reporting not required for this SBIR Phase 2 project. |
| **Bibliography Type:** | Description: (Last Updated: ) |